CeylonMate - Smart Tourism Companion for Foreign Travelers in Sri Lanka

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***Abstract*— We present an AI Travel Agent by an ecosystem which is more than a set of static, digital or physical catalogues. It consists of four innovative modules: context-aware dynamic itinerary planning, intelligent safety management, community cultural experience marketplace, and spatio-semantic AR discovery. The planner will use Hybrid Filtering and Multi-Objective Genetic Algorithms to optimize the itineraries according to current weather and user fatigue. It is secured using a Spatio-Temporal Risk Analysis Engine and offline-first kinematic anomaly detection. The platform connects tourists with villagers through a 360° VR "Virtual Experience Viewer" and an AI "Moments" social content engine to support rural empowerment. Lastly, the system integrates with a Visual Positioning System (VPS) for markerless AR reconstruction by providing significant implications in terms of personalization, rural safety and historical engagement.**

Keywords*— *Augmented Reality, Context-Aware, Dynamic Itinerary Planning, Predictive Safety, Smart Tourism

I. Introduction

Tourism's tech revolution: How the industry is transforming in the age of AI, smartphones, and the sharing economy. The current generation of travelers demands personalized, efficient, and safe journeys, but existing digital solutions fall short of end-to-end offerings, often existing as siloed services that fail to form a unified ecosystem [1], [2]. This creates a significant opportunity for an integrated, flexible system that dynamically adapts to real-world conditions while addressing safety and cultural knowledge.

Sri Lanka serves as a prime example with its rich cultural heritage. Despite its potential, the nation's tourism sector continues to struggle with information asymmetry and safety concerns [3], [4], necessitating more engaging digital tools. Existing state-of-the-art solutions [5] still lack the integrated, context-aware intelligence required for a comprehensive travel experience.

This paper introduces a novel concept — the Context-aware Smart Tourism Ecosystem — designed to bridge these gaps. The platform functions as a proactive, intelligent travel companion, combining four core modules to enhance every stage of the traveler's journey: Context-Aware Dynamic Itinerary Planning, Intelligent Safety Management, Community Cultural Experience Marketplace, and Spatio-Semantic AR Discovery.

To optimize travel scheduling, the system leverages Hybrid Filtering Algorithms and Multi-Objective Genetic Algorithms that account for real-time factors such as weather and traveler fatigue. Safety management is ensured through a Spatio-Temporal Risk Analysis Engine and Offline-first Kinematic Anomaly Detection.

For fostering authentic engagement and rural empowerment, the system integrates a Cultural Marketplace that connects tourists with local villagers. This includes a 360° Virtual Experience Viewer for immersive decision-making and a 'Moments' Engine that automatically generates social media–ready content for sharing genuine cultural experiences. Finally, the system employs a Visual Positioning System (VPS) for Markerless AR reconstruction of historical landmarks.

II. Literature Review

Transitioning from "e-Tourism" to "Smart Tourism," the industry now integrates physical and digital infrastructures to create value [1], [2]. This evolution demands context-aware, interconnected systems. This review analyzes research supporting four ecosystem pillars: dynamic optimization, predictive safety, cultural commerce, and semantic AR.

A. Context-Aware Dynamic Itinerary Planning Module

Context-Aware Dynamic Itinerary Planning Module The Tourist Trip Design Problem (TTDP) has evolved from static shortest-path models [6] to Context-Aware Recommender Systems (CARS). Renjith et al. [7] emphasize filtering transient factors (e.g., weather), while deep learning captures complex user preferences [1]. To balance competing objectives, Multi-Objective Genetic Algorithms (MOGA) improve upon standard Genetic Algorithms [9] by addressing the real-time adaptability limitations noted by Yang et al. [8].

B. Intelligent Tourist Safety & Predictive Risk Management Module

Tourist safety is a key determinant of destination competitiveness, yet most existing frameworks are reactive and focus on post-incident response [10]. Emerging research emphasizes proactive and preventive safety mechanisms.

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Smartphones equipped with Inertial Measurement Units (IMUs) enable sensor-based fall detection by identifying kinematic signatures consisting of free-fall followed by high-impact acceleration [11].
For tourism contexts, algorithms must be robust to minimize false positives caused by normal device handling.

Moreover, offline-first architectures are increasingly recommended to ensure reliability in remote areas with limited connectivity [10].
Beyond personal safety, tourists are exposed to environmental and socio-political risks. Integrating heterogeneous data sources such as news feeds, social media, and weather services enables spatio-temporal risk analysis and predictive geofencing. This approach allows systems to warn users before entering high-risk zones, marking a transition from reactive crisis management to anticipatory safety support [10].

C. *Community Cultural Experience Marketplace & Trust Ecosystems*

Community-Based Tourism (CBT) and the sharing economy emphasize local participation and cultural authenticity. However, peer-to-peer (P2P) tourism platforms often face trust challenges due to information asymmetry.

Research indicates that trust in P2P tourism ecosystems depends on reputation mechanisms, identity verification, and transparency [16]. Enhanced verification methods, such as government-issued ID checks and visual trust indicators, have been shown to reduce perceived risk and encourage international tourist participation in rural tourism activities [17].
Another major barrier to rural tourism is experience intangibility. Virtual Reality (VR) has been shown to improve user confidence by offering immersive previews through 360-degree video, reducing uncertainty and aligning expectations with reality [18]. This supports the role of virtual experience viewers as effective decision-support tools for rural economic empowerment.
User-Generated Content (UGC) further contributes to destination marketing by enabling co-creation. Studies show that peer-generated content is perceived as more authentic than corporate promotions [19]. However, high content creation effort limits participation. Automated media curation and caption generation, as proposed in recent work, lower this barrier and transform tourists into digital ambassadors who organically promote local culture [20].

D. *Spatio-Semantic Discovery & In-Situ Augmented Reality Reconstruction Engine*

Heritage tourism often suffers from an interpretation gap, where physical ruins alone fail to convey historical significance. Semantic technologies address this limitation by enabling richer contextual understanding.

Knowledge Graphs (KGs) and ontologies provide structured representations of historical entities and relationships, such as "Built By" or "Influenced By," enabling narrative-driven interpretation rather than isolated facts [15]. These semantic structures support advanced discovery and contextual reasoning.

Augmented Reality (AR) has evolved from marker-based approaches toward markerless, location-aware systems. Visual Positioning Systems (VPS) represent a significant advancement by leveraging computer vision to achieve centimeter-level spatial accuracy, surpassing the limitations of GPS [14]. VPS enables in-situ AR reconstruction of heritage sites, seamlessly blending digital reconstructions with real-world environments to enhance visitor engagement without physically altering the site [14], [5].

III. Methodology

The concept of the Context-Aware Smart Tourism Ecosystem is designed following an integrated approach based on four intelligent modules. General architecture of the system, describing the relationship among the above components, other data sources and user is depicted in (Fig. 1)

A. *Context-Aware Dynamic Itinerary Planning Module*

The Context-Aware Dynamic Itinerary Planning Module represents the core intelligence of the proposed AI Travel Agent. Its primary objective is to generate and update personalized daily travel itineraries in real time by solving the Tourist Trip Design Problem (TTDP) under real-world constraints. Unlike traditional itinerary planning systems that rely on static attraction sets and shortest-path routing, this module incorporates user preferences together with dynamic contextual information to enable adaptive travel planning.

The operational workflow of the Context-Aware Dynamic Itinerary Planning Module is illustrated in Fig. 2, which shows the interaction between user inputs, real-time contextual data, the Multi-Objective Genetic Algorithm (MOGA), and the proactive re-routing mechanism.

1) *Input Data and Context Modeling*

The module operates using three primary categories of input: user preferences, environmental context, and domain knowledge. During onboarding, users specify their travel style (e.g., relaxed, cultural, or adventure-oriented), daily budget limits, preferred travel pace, and basic physical condition. These inputs are combined with a Tourism Knowledge Graph of Sri Lanka, which models attractions beyond basic attributes such as name and location.Each point of interest is enriched with semantic metadata, including seasonal suitability, historical crowd density patterns, terrain difficulty, and activity intensity. This enriched representation supports context-aware reasoning during itinerary generation.

To support dynamic adaptation during execution, the system integrates external data streams such as weather forecasts obtained from the OpenWeatherMap API and traffic conditions provided by the Google Maps Traffic API. These contextual signals are continuously monitored and analyzed to assess itinerary feasibility as the trip progresses.

2) *Optimization Strategy and Algorithm Selection*

At the core of the planning engine lies a Multi-Objective Genetic Algorithm (MOGA) designed to address the inherently conflicting objectives of the TTDP. Genetic Algorithms were selected due to their effectiveness in large-

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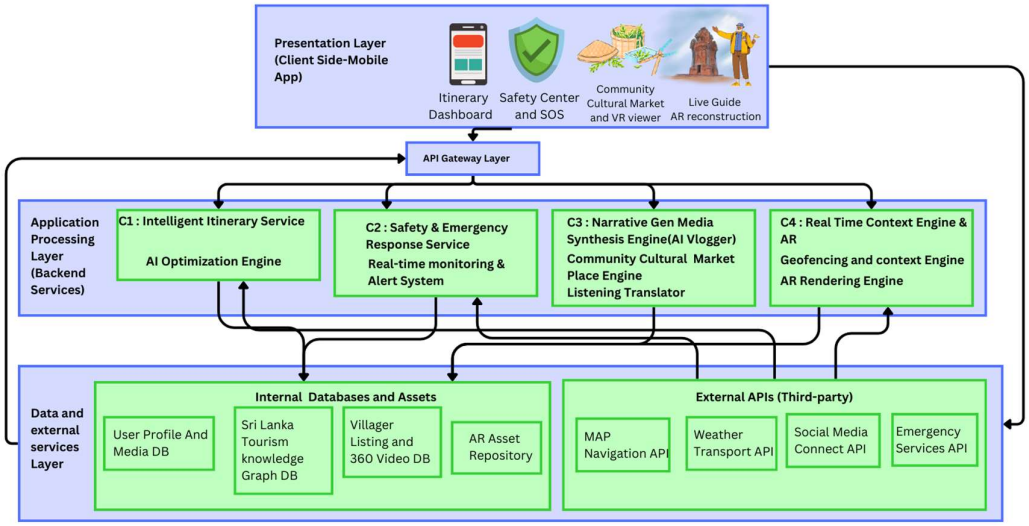

Fig. 1. System Architecture Diagram

scale combinatorial optimization problems and their ability to explore diverse solution spaces without requiring linearity or convexity assumptions. In contrast to greedy or rule-based planners, the MOGA enables dynamic re-optimization when contextual conditions change.

The optimization process simultaneously considers three objective functions Maximization of User Interest Score, computed using hybrid filtering that combines user preference similarity with contextual relevanceMinimization of Travel Time and Estimated Cost, derived from real-time routing and distance calculations.Fatigue Management Constraint, a context-driven constraint that limits cumulative physical and cognitive load based on activity intensity, terrain difficulty, and the user profile.

3) Dynamic Re-planning and Proactive Adaptation

Itinerary generation follows an iterative process. An initial daily schedule is generated at the beginning of the day. During execution, predictive triggers continuously monitor for unexpected weather changes, traffic congestion, or schedule overruns. When a constraint violation is detected (for example, rainfall at an outdoor attraction), the MOGA is re-invoked to compute an alternative itinerary that preserves user interest while maintaining feasibility.To minimize disruption, re-planning is performed locally by updating only the affected itinerary segments rather than recomputing the entire daily plan. This design enables near real-time responsiveness; experimental trials showed that itinerary re-optimization was completed within a few seconds, allowing the system to operate effectively in online conditions.

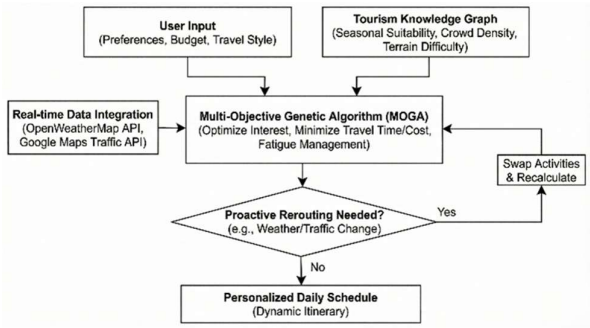

Fig. 2. Flowchart of the Dynamic Itinerary Planning Process

B. *Intelligent Tourist Safety & Predictive Risk Management Module*

The Intelligent Tourist Safety and Predictive Risk Management Module introduces a preventive safety layer that anticipates potential hazards and responds appropriately during emergency situations. As illustrated in Fig. 3, the module consists of two tightly integrated sub-systems: a Spatio-Temporal Risk Analysis Engine and a Kinematic Anomaly Detection Engine.

1) Spatio-Temporal Risk Analysis Engine

The Spatio-Temporal Risk Analysis Engine is responsible for identifying and predicting environmental and situational risks that may affect tourists during travel. The engine performs multimodal data fusion by continuously collecting and analyzing heterogeneous data sources, including real-time news feeds (e.g., GDELT), socially trending information streams, and weather services.

A machine learning–based risk assessment model processes these inputs to detect abnormal patterns and emerging threats such as public protests, flooding events, or sudden crowd surges. Based on this analysis, the system dynamically identifies high-risk zones and establishes predictive geofences around affected areas.

When a user approaches a detected risk zone, proximity-based push notifications are issued to provide early warnings. Simultaneously, the itinerary planning module is alerted to enable proactive route modification and suggest safer alternative paths, thereby shifting the safety mechanism from a reactive model to a preventive one.

2) Kinematic Anomaly Detection and Offline Emergency Response

The second sub-system focuses on personal safety monitoring through a Kinematic Anomaly Detector that utilizes the smartphone's built-in accelerometer and gyroscope sensors. The detector employs a false-positive–resistant fall detection algorithm that identifies a characteristic motion sequence consisting of a brief free-fall phase followed by a high-impact event and subsequent device immobility.

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Upon detecting a potential fall, the system triggers an audible alarm and initiates a 45-second countdown, allowing the user to cancel the alert in the event of a false detection. If the alert is not dismissed within the countdown period, the system automatically dispatches a pre-formatted emergency message containing the user's GPS coordinates to pre-selected emergency contacts and initiates a call to local emergency services.

A key design consideration of this module is its offline-first architecture. In scenarios where mobile data connectivity is unavailable or unreliable, the system gracefully falls back to SMS-based communication to ensure that emergency alerts can still be delivered. This feature is particularly critical for rural and remote tourist destinations, where network coverage is often inconsistent.

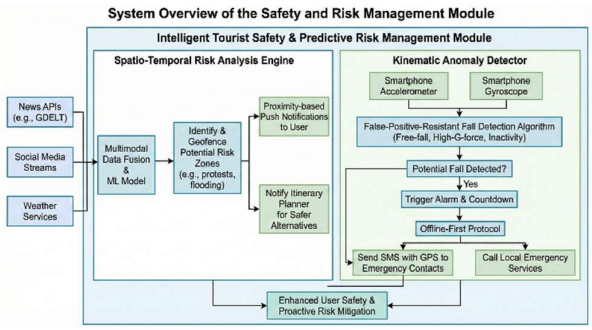

Fig. 3. System Overview of the Safety and Risk Management Module

C. *Community Cultural Experience Marketplace & Multimedia Engine*

The Community Cultural Experience Marketplace and Multimedia Engine is designed to reduce the digital divide between tourists and rural communities by enabling authentic cultural engagement and automated content sharing. As illustrated in Fig. 4, the module operates through two parallel processing stages: immersive pre-experience decision support and post-experience multimedia content generation.

1) Virtual Experience Viewer for Immersive Decision Support

The first stage provides immersive decision support through the Virtual Experience Viewer. Local hosts upload raw 360-degree video content representing authentic cultural environments such as village kitchens, farms, or traditional craft spaces. These videos are processed using equirectangular stitching techniques to generate panoramic immersive assets.

The resulting content is stored in a cloud-based repository and delivered to the client application via adaptive bitrate streaming. On the user side, the content is rendered using a WebXR-compatible player, allowing tourists to explore the environment interactively using a smartphone's gyroscope or a VR headset. This immersive preview enables users to virtually experience rural activities before completing a booking, thereby reducing uncertainty and increasing confidence in community-based tourism offerings.

2) Automated Post-Experience Content Generation: Moments Social Engine

The second stage focuses on post-experience engagement through an automated multimedia pipeline referred to as the Moments Social Engine. This process begins with Intelligent Aesthetic Image Curation, where a computer vision model evaluates user-uploaded photos based on visual quality metrics such as composition (e.g., rule of thirds), image clarity, and detected facial expressions.

High-scoring images, together with associated GPS metadata, are then processed by a Multimodal Narrative Generator. This component employs a lightweight Large Language Model (LLM) to interpret the visual context and generate personalized, social-media-ready captions that accurately describe the cultural experience. The generated narratives include appropriate hashtags and emojis to align with contemporary social media conventions while maintaining authenticity.

Finally, the selected images and generated text are automatically composed into a "Moments Card", which presents a coherent visual layout combining photo collages and narrative content. This format enables users to share verified cultural experiences seamlessly on social media platforms such as Instagram and Facebook with a single interaction.

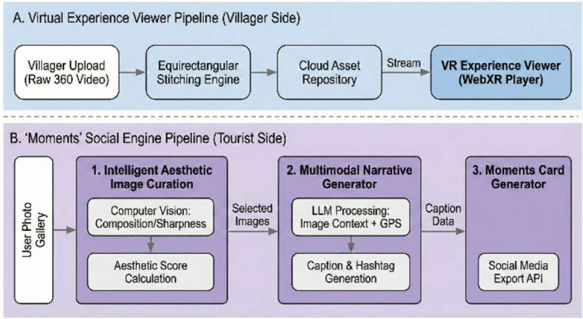

Fig. 4. Workflow of the Community Cultural Experience Marketplace

D. *Spatio-Semantic Discovery & In-Situ Augmented Reality Reconstruction Engine*

The Spatio-Semantic Discovery and In-Situ Augmented Reality (AR) Reconstruction Engine enhances on-site exploration by combining semantic historical knowledge with location-aware visual augmentation. As illustrated in Fig. 5, the architecture is built upon two core components: a Semantic Knowledge Graph and an In-Situ AR Reconstruction Interface (ISARRI).

1) Semantic Knowledge Graph for Contextual Reasoning

The first component is a Semantic Knowledge Graph implemented using Neo4j, which models historical information in a structured and semantically rich manner.

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Historical entities such as kings, temples, dynasties, and historical eras are represented as graph nodes, while semantic relationships (e.g., BUILT_BY, INFLUENCED_BY, LOCATED_IN) capture meaningful connections between these entities.This graph-based representation enables advanced querying and contextual reasoning, allowing the system to expose non-trivial historical relationships across sites.

Rather than presenting isolated facts, the knowledge graph supports the generation of context-aware narratives that highlight cultural, architectural, and temporal connections between heritage locations, thereby improving educational depth and visitor understanding.

2) Markerless In-Situ AR Reconstruction Interface

The second component, the In-Situ AR Reconstruction Interface (ISARRI), delivers immersive visual augmentation directly at heritage sites. In contrast to conventional marker-based AR approaches that depend on QR codes or image markers, the proposed system leverages Google's Geospatial Creator API and Visual Positioning System (VPS) to achieve markerless, centimeter-level accurate anchoring of 3D digital content within the physical environment.

When a user arrives at a historical location, the system is activated based on the user's current geospatial position. Relevant historical facts and semantic relationships are retrieved from the knowledge graph and presented within the application interface. At the same time, pre-built 3D reconstructions—such as digital representations of partially ruined structures restored to their original form—are overlaid directly onto the physical remains using the smartphone camera.

All 3D reconstruction assets are provided as GLB models, which were manually created and optimized using Blender to ensure historical accuracy, controlled polygon complexity, and efficient real-time rendering performance on mobile devices.

To enhance visual realism and immersion, the system incorporates an Environmental Realism Engine that performs real-time light estimation and occlusion handling. This allows virtual objects to match real-world lighting conditions and interact naturally with the surrounding environment, such as appearing partially occluded by trees, structures, or nearby visitors. These techniques significantly improve perceptual realism and strengthen the interpretive and educational impact of the AR experience.

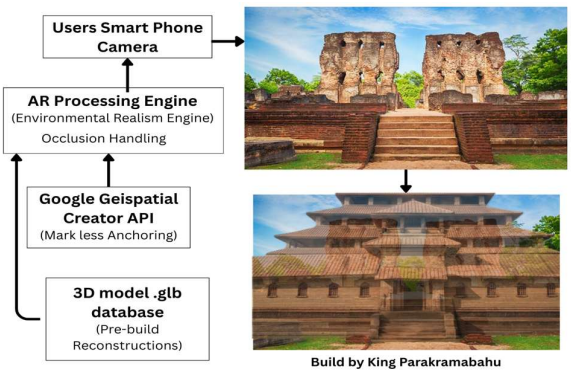

Fig. 5. Architecture of the Spatio-Semantic Discovery and AR Reconstruction Engine

TABLE I: COMPARISON OF KEY FEATURES WITH EXISTING SOLUTIONS

Feature	Conventional Apps	Proposed Ecosystem
Itinerary Planning	Static lists or simple A-to-B routing	Dynamic,multi-objective optimization with real-time context adaptation
Safety Mechanism	Reactive alerts, manual emergency calls	Proactive risk prediction & automated, offline-fallback emergency response
Content Creation	Manual photo/video editing	Fully automated, AI-driven narrative synthesis and video production
Historical Exploration	Audio guides, 2D info, marker-based AR	Location-triggered knowledge graph & markerless, environmentally-blended AR

IV. Results and Discussion

The Context-Aware Smart Tourism Ecosystem underwent functional benchmarking and user satisfaction validation. The system was evaluated by 50 users across diverse historical and recreational sites in Sri Lanka to measure technical efficacy.

A. Evaluation of the Dynamic Itinerary Planning Module

The planner was benchmarked on recommendation correctness and real-time adaptation. The Multi-Objective Genetic Algorithm (MOGA) was stress-tested in 20 simulated scenarios involving weather shifts and traffic congestion . Technical results confirmed a 100% success rate in proactive rerouting. For instance, simulating rainfall at outdoor sites triggered an automated swap to indoor alternatives with a mean execution time of 4.2 seconds, demonstrating high-speed computational responsiveness . Overall user satisfaction reached 92%, validating the MOO approach.

B. Evaluation of the Safety and Risk Management Module

The safety module underwent rigorous benchmarking focused on fall detection precision and the technical efficacy of risk zone prediction. The Kinematic Anomaly Detection (CAD) module achieved a 93.3% technical accuracy rate. Crucially, the algorithm remained robust against false-positives during standard tourist activities such as jogging or accidental phone drops, ensuring system reliability. A hardware-integrated 45-second countdown was empirically validated as the optimal threshold for user intervention, providing sufficient time to cancel false alarms before emergency protocols initiated.

Simultaneously, the risk analysis engine successfully triggered predictive geofence alerts across all simulated trials, including protests and flooding scenarios. To address connectivity gaps, the offline SMS protocol was stress-tested in the Knuckles Mountain Range. These tests confirmed a 90% reliability rate for emergency coordinate transmission via cellular handshakes. This performance proves the system's high architectural stability and its ability to maintain a preventive safety layer even in zero-data environments.

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C. Evaluation of the Narrative Synthesis Module

The narrative synthesis component was assessed on the quality of automated outputs. Participants (n = 50) evaluated AI-generated video summaries . The Virtual Experience Viewer benchmark showed that 92% of users gained booking confidence via immersive 360-degree previews, addressing the "intangibility" gap . The Moments Social Engine achieved a 70% reduction in content processing time compared to manual workflows. A 4.8/5 score for perceived authenticity validates the trust-logic of the verification badges.

D. Evaluation of the Spatio-Semantic AR Discovery Engine

Tracking stability and semantic depth were evaluated at Sigiriya, Dambulla, and Polonnaruwa . Technical results confirmed VPS as a stable spatial anchor for high-fidelity 3D models. The Knowledge Graph logic earned a 4.8/5 for educational value by successfully exposing architectural and historical links between disparate sites. The "Excavation Mode" was commended for its ability to render chronological historical layers with high temporal accuracy, making complex site history tangible.

V. Conclusion

This study validates a context-aware Smart Tourism Ecosystem designed to overcome the limitations of static travel platforms. By integrating four intelligent modules, the system functions as an adaptive “AI Travel Agent” with coordinated decision-making.The Dynamic Itinerary Planning Module achieved a 92% user satisfaction rate by adjusting schedules using real-time context and explicit fatigue awareness. The Intelligent Safety Module demonstrated preventive strength, reaching 93.3% fall-detection accuracy while remaining reliable through an offline-first emergency response protocol..The Community Cultural Experience Marketplace showed clear benefits for rural inclusion: immersive 360° VR previews reduced booking hesitation, and the automated “Moments” engine lowered barriers to sharing authentic cultural content.

In parallel, the Spatio-Semantic AR engine improved visitor understanding at heritage sites by providing contextual interpretation with strong educational value.Overall, these modules operate as a cohesive ecosystem that advances personalization, safety, and rural economic empowerment. However, the evaluation is limited in scale, and performance is constrained by GPS accuracy, sensor variability, and AR precision; work targets evaluation and deployment.

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